

Md. Ajmain Istiak Apon

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EDUCATION

University of Dhaka

B.Sc. in Robotics and Mechatronics Engineering

Relevant Coursework: Machine Learning, Artificial Intelligence, Intelligent Systems and Robotics, Digital Image Processing and Robot Vision, Digital Signal Processing, Linear Algebra, Statistics for Engineers.

Dhaka, Bangladesh

Expected Oct 2026

EXPERIENCE

Undergraduate Research Assistant

2024 – 2025

University of Dhaka — Funded Research Project: Stomatal Phenotyping — paper under review

- Built a **506-image two-class instance-segmentation benchmark** (6,605 guard-cell-complex and pore annotations) from chickpea leaf microscopy; annotated in **Roboflow**, exported in COCO format with locked train/val/test splits (359/99/48 images).
- Benchmarked **YOLOv8s-seg**, **YOLO26s-seg**, and **Mask R-CNN** on identical splits; YOLO26s-seg achieved **0.707 box mAP@0.50** and **0.638 mask mAP@0.50**, Mask R-CNN led in mask **mAP@0.50:0.95 (0.428)** on the locked 48-image held-out test.
- Automated stomatal density and pore-index extraction from predicted masks, validated against manual counts ($p < 0.05$); performed PCA and t -test analyses linking morpho-physiological and stomatal traits to drought-stress responses in *C. arietinum*.

RESEARCH & PROJECTS

Senior Thesis — GenMem

Jan 2026 – Present

Auditing Predicted-Memorability Guidance in Text-to-Image Diffusion — under faculty supervision

- Implemented **GenMem**, a training-free guidance method that differentiates a pretrained memorability predictor (**AMNet**) through the **DDIM** sampling trajectory of **Stable Diffusion**, leaving all model weights frozen.
- On **DrawBench** ($n=200$): held-out memorability rose **+3.42%** (ResMem) and **+2.72%** (ViTMem), both $p < 10^{-8}$, with no significant CLIP-fidelity loss; replicated on **COCO** ($n=200$) with equal or larger effect sizes.
- Validated causality via reverse-guidance symmetry and a null random-critic control; characterized proxy over-optimization in SDXL and mapped a three-regime dose-response curve across an 8-point guidance-weight sweep.

Self-Supervised Learning for Label-Efficient Medical CT Segmentation

🔗 ajmainapon/SSLR

First-author paper — under review at WACV 2027 (Algorithms Track)

Jan 2026 – Present

- Adapted **I-JEPA** to 2.5D CT slice triplets with a **ViT-B/16** backbone and EMA target encoder on **1,228 unlabeled TotalSegmentator volumes**; LOO ablation showed natural-image design choices invert on medical CT—narrow slice gap yielded the **A3** backbone, improving linear-probe Dice by **44–124%**.
- **A3** features significantly outperform random init under frozen and fine-tuned probing at $N=20/50/100$ labels (Welch t : $p=0.017$ at $N=20$, $p<0.001$ at $N=50$); achieved **67% recovery** of **nnU-Net** supervised performance at $N=50$, liver reaching **90%**; seed variance reduced **1.3–2.2×**.
- Validated **cross-dataset** transfer on AMOS22-CT (**+158–260%** over random init) and **cross-modality** CT→MR (**+41–48%** at $N=20/50$); reproduced collapsing SSL gain on 3D **SwinUNETR**, confirming cross-architecture robustness. *Stack:* PyTorch, timm, MONAI, Nibabel, CUDA.

REST API with JWT Auth (FastAPI)

🔗 ajmainapon/FASTAPI-Project

Personal Project

Apr 2026

- Built a multi-router FastAPI service for an articles/authors blog backend with **SQLAlchemy** ORM and **Pydantic v2** request/response schemas.
- Implemented **OAuth2 password-flow** login issuing **JWT bearer tokens** (bcrypt password hashing via passlib); structured the app with dependency-injected DB sessions and per-resource routers.

TECHNICAL SKILLS

Languages Python, C++, C, SQL, JavaScript

ML / CV PyTorch, Ultralytics, timm, torchvision, TensorFlow, Scikit-learn, Hugging Face, OpenCV, MONAI, Roboflow

LLM / API LangChain, LangGraph, FastAPI, SQLAlchemy, Pydantic, OAuth2 / JWT

COMPETITIVE PROGRAMMING

Codeforces (fynm): 500+ problems solved across dynamic programming, graph theory, greedy, and advanced data structures.

LEADERSHIP

• **President**, Robotics & Mechatronics Student Club, University of Dhaka

Oct 2025 – Present

• **Program Coordinator**, IEEE Robotics & Automation Society, University of Dhaka

Mar 2024 – Apr 2025